MILITARY SPECIFICATION

COATING SYSTEM, THERMAL INSULATION, INTUMESCENT

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope. This specification covers the requirements for a two coat thermal insulating system for coating the exterior surfaces of aircraft ordnance to provide protection from high-flux thermal environments.

1.2 Classification. The coatings of the thermal insulation system furnished under this specification shall consist of the following types as specified (see 6.2):

Type I - Base coat (Two-component, pigmented epoxy resin)

Type II - Topcoat (Two-component, pigmented epoxy resin)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-250/5	Aluminum Alloy, Alclad 2024, Plate and Sheet
QQ-A-250/11	Aluminum Alloy 6061, Plate and Sheet
PPP-D-729	Drums, Metal, 55 Gallon (For Shipment of Non-corrosive Materials).

FSC 8010

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SPECIFICATIONS

Federal (Continued)

PPP-P-704	Pail, Shipping, Steel (1 Through 12 Gallon)

Military

MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-P-23377	Primer Coating, Epoxy - Polyamide, Chemical and Solvent Resistant
MIL-C-81706	Chemical Conversion Materials For Coating Aluminum and Aluminum Alloys

STANDARDS

Federal

Fed. Test Method Std. No. 141	Paint, Varnish, Lacquer and Related Materials, Methods of Inspection, Sampling and Testing
Fed. Test Method Std. No. 406	Plastic, Methods of Testing
FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners, or Pallet Type Base

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STANDARDS

Military (Continued)

MIL-STD-810	Environmental Test Methods

DRAWINGS

Department of the Navy - Naval Air Systems Command

4902292	- Launcher, Rocket LAU-10D/A

PUBLICATIONS

Department of the Navy

WR-50	Warhead Safety Tests, Minimum For Air, Surface and Underwater Launched Weapons

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM D 1014	Conducting Exterior Exposure Tests of Paints on Steel
ASTM D 1748	Rust Protection by Metal Preservatives in the Humidity Cabinet
ASTM D 2073	Total, Primary, Secondary, and Tertiary Amine Values of Fatty Amines, Aminoamines and Diamines by referee Potentiometric Method

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(Application for copies of ASTM Standards should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Qualification. The coating system furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.3 and 6.3). Any change in the formulation of a qualified product will necessitate its requalification. The material supplied under the contract shall be identical, within manufacturing tolerances, to the product receiving qualification.

3.2 Composition. The coating system shall consist of two separate coatings; Type I, the base coat, and Type II, the topcoat. Each coating furnished under this specification shall consist of two separately packaged components. Component I of each coating shall consist of a pigmented solventless sprayable epoxy resin. Component II of Type I coating shall consist of a modified polyamide curing agent and Component II of Type II coating shall consist of a modified polyamine curing agent.

3.2.1 Formulation. When Component I and Component II of each coating are mixed in the following ratios, base coat and topcoat compounds meeting the requirements of this specification shall result:

Type I (Base Coat)	Parts by Weight
Component I (Resin)	4
Component II (Curing agent)	1

Type II (Topcoat)	Parts by Weight
Component I (Resin)	9
Component II (Curing agent)	1

3.3 Toxicity. The material shall have no adverse effect on the health of personnel when used under the conditions specified in 5.3.1. Questions pertinent to this effect shall be referred by the procuring activity to the appropriate department medical service who will act as an advisor to the procuring activity.

3.4 Component requirements.

3.4.1 Type I, Component I (Resin). Type I, Component I, shall conform to the requirements specified in Table I, 3.4.5 and 3.4.6.

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TABLE I

TYPE I, COMPONENT I REQUIREMENTS

Characteristics	Requirements		Test Paragraph
	Minimum	Maximum	
Viscosity, at 120 ±2°F, Centipoises	100,000	200,000	4.6.1.1
Solids Content, percent	29.5	33.5	4.6.1.2
Weight per gallon, pounds	10.5	10.9	4.6.1.3
Epoxide equivalent	470	535	4.6.1.4

3.4.2 Type II, Component I (Resin). Type II, Component I, shall conform to the requirements specified in Table II, 3.4.5 and 3.4.6.

TABLE II

TYPE II, COMPONENT I REQUIREMENTS

Characteristics	Requirements		Test Paragraph
	Minimum	Maximum	
Viscosity, at 73 ±2°F, Centipoises	20,000	45,000	4.6.1.1
Solids content, percent	17.5	21.5	4.6.1.2
Weight per gallon, pounds	10.0	10.5	4.6.1.3
Epoxide equivalent	395	450	4.6.1.4

3.4.3 Type I, Component II (Curing agent). Type I, Component II, shall conform to the requirements specified in Table III, 3.4.5 and 3.4.6.

TABLE III.

TYPE I, COMPONENT II REQUIREMENTS

Characteristics	Requirement		Test Paragraph
	Minimum	Maximum	
Total Amine Value	340	405	4.6.2.1
Viscosity at 150 ±2°F Centipoises	1,000	3,000	4.6.2.2
Weight per gallon, pounds	7.7	8.2	4.6.2.3

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3.4.4 Type II, Component II (Curing agent). Type II, Component II, shall conform to the requirements specified in Table IV, 3.4.5, and 3.4.6.

TABLE IV.

TYPE II, COMPONENT II REQUIREMENTS

Characteristics	Requirements		Test Paragraph
	Minimum	Maximum	
Total Amine Value	935	1600	4.6.2.1
Viscosity at 77 ±2°F Centipoises	5,000	13,000	4.6.2.2
Weight per gallon, pounds	8.8	9.2	4.6.2.3

3.4.5 Condition in container. Type I and II component materials packaged in closed containers shall be capable of being easily mixed to a smooth homogeneous condition. There shall be no trace of pigment flotation, grit, seeds, skins or lumps and no evidence of hard or objectionable settling which cannot be readily dispersed.

3.4.6 Storage stability. The previously unopened packaged components shall meet the following requirements after being stored a minimum of 6 months at 65 ±25° from the date of manufacture (see 4.6.4):

Component	Requirement
Type I, Component I	Table I
Type II, Component I	Table II
Type I, Component II	Table III
Type II, Component II	Table IV

3.4.6.1 Accelerated stability. A full unopened can of Type I and II, Component I material shall meet the requirements of Table I and II respectively and the condition in container requirements, 3.4.5, when exposed as specified in 4.6.4.1.

3.5 Application, Type I and II. When components I and II of Type I coating and components I and II of Type II are admixed separately at the ratio specified in 3.2.1, the coating compounds shall be freely working products when applied as specified in 4.6.5.

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3.6 Coating properties.

3.6.1 Thermal efficiency. The thermal efficiency of the coating system shall be a minimum of 1.2 seconds per mil to reach a backface temperature of 260°C (500°F) and 2.0 seconds per mil to reach a backface temperature of 538°C (1000°F) (see 4.6.6) before and after being subjected to the humidity test of 4.6.14.

3.6.2 Tensile properties (Type I). The minimum tensile properties of the coating compound when tested as specified in 4.6.7 shall be as follows:

TEMPERATURE (°F)	ULTIMATE STRENGTH (psi)	ELONGATION AT BREAK (percent)
-40	2,000	.7
75	900	5.0
140	250	6.0

3.6.3 Color.

3.6.3.1 Type I. Type I components shall be supplied in the manufactured color. When admixed in the ratio specified in 3.2.1, the resultant color of the coating shall be off-white.

3.6.3.2 Type II. The color and gloss of Type II cured coating shall match color number 17875 of FED-STD-595 (see 4.6.8). By photoelectric tristimulus, the minimum values shall be as follows:

A value	75
B value	75
G value	75

3.6.4 Hardness (Type I). The cured Type I coating shall have a minimum Shore A hardness of 85 (see 4.6.9).

3.6.5 Adhesion. Panels coated as specified in 4.6.10 when bent around a mandrel 1 +0.003 inch in diameter, shall not exhibit adhesive failure. Cracking of the coatings is acceptable and shall not be cause for rejection.

3.7 Resistance properties.

3.7.1 Lubricating oil resistance. The cured coating system shall withstand immersion in oil for a period of 4 hours at a temperature of 250 ±4°F. Four hours after removal, the coating shall show no blistering, softening, or other defects. Discoloration shall not be cause for rejection (see 4.6.11).

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3.7.2 Fungus resistance. The cured coating system shall be non-nutrient after subjection to the fungus test specified in 4.6.12.

3.7.3 Impact resistance. The cured coating system when tested as specified in 4.6.13 shall require more than 75 inch-pounds on direct (coated surface) impact and 7 inch-pounds minimum on reverse impact before cracking or flaking is evident.

3.7.4 Humidity resistance. The cured coating system on the panels, specified in 4.6.14 shall have a minimum Shore A hardness of 75 upon removal of the panels from the cabinet.

3.7.5 Resistance to temperature shock. There shall be no evidence of cracking, loss of adhesion, or film deterioration of the cured coating system when tested as specified in 4.6.15.

3.7.6 Weather resistance. There shall be no evidence of cracking or loss of adhesion of the coating after the one year exposure period specified in 4.6.16. After exposure, the coated panels shall meet the thermal efficiency requirements of 3.6.1, except that subjection of the panels to the humidity test shall not be required.

3.8 Environmental performance requirements. LAU-10D/A rocket launchers conforming to Naval Air Systems Command Drawing No. 4902292 shall be coated with the thermal insulation coating system specified in 4.5.2. The dry film thickness of the coating system shall be 170 ±15 mils. The rocket launchers shall then be tested for conformance to the requirements of 3.8.1 and 3.9.

3.8.1 Environmental performance. The coating system shall pass all the environmental tests specified in 4.6.17 without evidence of loss of adhesion, excessive cracking, blistering or flaking.

3.9 Fast cook-off. LAU-10D/A launchers coated as specified in 3.8 shall be subjected to the fast cook-off test specified in 4.6.18. The cook-off time shall be an average of 9 minutes minimum.

3.10 Workmanship. Type I and II coatings shall be processed in a manner that will produce the highest quality material necessary to meet the requirements of this specification. Each component shall be uniform in quality and consistency after stirring, and free of agglomerates or foreign particles. There shall be no other defects present that would render the end product unsuitable for its intended purpose.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection of the coating system shall be classified as follows:

a. Qualification inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 Qualification.

4.3.1 Inspection. The qualification inspection of the coating system shall consist of examinations and tests for all the requirements of this specification, and a review for approval of the supplier's report of tests.

4.3.1.1 Retention of qualification. In order to retain qualification of products approved for listing on the Qualified Products List (QPL), the manufacturer shall verify by certification to the qualifying activity that his product(s) comply with the requirements of this specification. Unless otherwise specified by the qualifying activity, the time of periodic verification by certification shall be in two-year intervals from the date of original qualification.

4.3.2 Samples. Qualification samples shall consist of wet samples and test specimens necessary for performing the tests specified under 4.6. Wet samples shall consist of one gallon of Component I of Type I and II and one pint of Component II of Type I and II coating. In addition, the manufacturer shall submit wet (one pint) and dry (one pound) samples of each ingredient material used in the formulation of Types I and II coatings. The samples shall be representative of the materials to be used during production. The wet samples and test specimens shall be forwarded to the Commanding Officer, U.S. Naval Air Development Center, Warminster, Pennsylvania 18974, Attention: Code 30223. The samples shall be legibly marked with the following information:

a. Coating System, Thermal Insulation, Intumescent.

b. Type (I or II, as applicable).

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c. Component (I or II, as applicable).

d. Specification MIL-C-81946(AS).

e. Name and address of manufacturer.

f. Location and identity of the plant which produced the samples.

g. Manufacturer's identification.

h. Submitted by (name) (date) for qualification tests in accordance with MIL-C-81946(AS), under authorization of (reference authorizing letter) (see 6.3).

4.3.2.1 Test samples for qualification shall be submitted with a notarized report of all tests specified herein except for weather resistance and storage stability. Report of tests shall be as specified in 4.4.1.

4.3.3 Qualification tests. The qualification tests of the coating system shall consist of all the tests specified under 4.6 "Test methods".

4.4 Quality conformance inspection. Quality conformance inspection shall consist of all examinations of this specification and the tests specified in Table V under the column titled "Required Testing for Quality Conformance Inspection".

4.4.1 Report of tests. The manufacturer shall submit notarized test reports to the Government representative, in accordance with Method 1031 of Fed. Test Method Std. No. 141, for each batch showing the results including numerical values when applicable of all the tests specified herein. Reports of numerical tests as satisfactory are not acceptable.

4.4.2 Sampling.

4.4.2.1 Lot. An inspection lot shall consist of all the coating compounds of the coating system from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.5).

4.4.2.2 Sampling for tests. Unless otherwise specified, sampling for quality conformance tests shall be in accordance with Method 1021 of Fed. Test Method Std. No. 141. The samples selected shall be sufficient in size to conduct all required tests.

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4.4.2.3 Sampling for visual inspection of filled containers. A random sample of filled containers shall be selected in accordance with MIL-STD-105, Inspection Level S-4, Acceptable Quality Level (AQL) of 2.5 percent defective for examination as specified in 4.4.3.2, 4.4.3.3 and 4.4.4.

4.4.3 Inspection procedures.

4.4.3.1 Tests. The quality conformance tests of the components and coating system shall consist of all the tests specified in Table V and such other tests deemed necessary to check the manufacturer's data. If a test specimen fails to meet any of the tests required by this specification, the lot represented by the sample shall be rejected.

4.4.3.1.1 In addition, the Government reserves the right to have any of the tests called for by this specification rerun at a Government laboratory at any time within 3 months from the date of manufacture, as attested by the date appearing on the label. Samples for these tests shall be taken from previously unopened containers which have been stored as specified in 4.6.4.

4.4.3.2 Examination of preparation for delivery. The samples selected in accordance with 4.4.2.3 shall be examined for packaging, packing, marking, leakage, and workmanship. Any package having one or more defects shall be rejected. If the number of defective packages exceeds the Acceptable Quality Level (AQL) of 2.5 percent defective in accordance with MIL-STD-105, the lot represented by the sample shall be rejected.

4.4.3.3 Examination for container fill. The size sample of 4.4.3.2 shall be adjusted to Inspection Level S-3 of MIL-STD-105 except that the sample size derived from MIL-STD-105 shall be rounded off to the next highest increment of five. At no time shall the sample size be less than five, unless the lot consists of less than five containers in which case all containers shall be used to determine the fill. The sample unit shall be one filled container. The lot shall be rejected if the average weight per container is less than that specified.

4.4.4 Examination of product. Conformance of the coating compound to the requirements for workmanship and any other requirements not covered by specific tests shall be determined by appropriate visual examination.

4.5 Test conditions. The laboratory testing conditions shall be in accordance with Fed. Test Method Std. No. 141, and as described herein.

4.5.1 Test panels. Except as otherwise specified herein, test panels shall be 6061 aluminum alloy conforming to QQ-A-250/11, condition T-6. Each panel shall be 0.040 by 4 by 12 inches in size and treated with materials conforming to MIL-C-81706 to produce coatings conforming to MIL-C-5541.

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4.5.1.1 Panel preparation. Epoxy-polyamide primer conforming to MIL-P-23377 shall be applied to the panels to a dry film thickness of 0.6 to 0.8 mils and cured 1 hour at 160° +4°F.

4.5.1.2 Application of coatings.

4.5.1.2.1 Type I. Components I and II shall be thoroughly mixed in the ratio as specified in 3.2.1 until a homogeneous color is obtained, and with a doctor blade, spread to a thickness of 0.060 ±.010 inch on the prepared surface of the test panel. Care shall be exercised to insure a uniform coating free from entrapped air bubbles or other film defects. The coated panels shall then be cured for 15 minutes at 150 ±2°F.

TABLE V

REQUIREMENT AND TEST METHOD INDEX

Item	Requirement Paragraph	Paragraph Giving Further Reference	Required Testing For Quality Conformance Inspection
Type I, Component I			
Viscosity	Table I	4.6.1.1	Yes
Solids Content	Table I	4.6.1.2	Yes
Weight per gallon	Table I	4.6.1.3	Yes
Epoxide Equivalent	Table I	4.6.1.4	Yes
Condition in container	3.4.5	4.6.3	Yes
Storage stability	3.4.6	4.6.4	--
Accelerated stability	3.4.6.1	4.6.4.1	--
Type II, Component I			
Viscosity	Table II	4.6.1.1	Yes
Solids Content	Table II	4.6.1.2	Yes
Weight per gallon	Table II	4.6.1.3	Yes
Epoxide Equivalent	Table II	4.6.1.4	Yes
Condition in container	3.4.5	4.6.3	Yes
Storage stability	3.4.6	4.6.4	--
Accelerated stability	3.4.6.1	4.6.4.1	--
Type I, Component II			
Total Amine Value	Table III	4.6.2.1	Yes
Viscosity	Table III	4.6.2.2	Yes
Weight per gallon	Table III	4.6.2.3	Yes
Condition in container	3.4.5	4.6.3	Yes
Storage stability	3.4.6	4.6.4	--

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TABLE V (Continued)

Item	Requirement Paragraph	Paragraph Giving Further Reference	Required Testing For Quality Conformance Inspection
Type II, Component II			
Total Amine Value	Table IV	4.6.2.1	Yes
Viscosity	Table IV	4.6.2.2	Yes
Weight per gallon	Table IV	4.6.2.3	Yes
Condition in container	3.4.5	4.6.3	Yes
Storage stability	3.4.6	4.6.4	--
Coating System			
Application	3.5	4.6.5	--
Thermal efficiency	3.6.1	4.6.6	--
Tensile properties	3.6.2	4.6.7	--
Color, Type I and Type II	3.6.3	4.6.8	Yes
Hardness	3.6.4	4.6.9	Yes
Adhesion	3.6.5	4.6.10	Yes
Lubricating oil resistance	3.7.1	4.6.11	Yes
Fungus resistance	3.7.2	4.6.12	--
Impact resistance	3.7.3	4.6.13	--
Humidity resistance	3.7.4	4.6.14	--
Resistance to temperature shock	3.7.5	4.6.15	--
Weather resistance	3.7.6	4.6.16	--
Environmental performance tests:	3.8 and 3.8.1	4.6.17	--
Temperature and humidity cycling	3.8.1	4.6.17.1	--
Transportation vibration and abrasion	3.8.1	4.6.17.2	--
Aircraft vibration	3.8.1	4.6.17.3	--
Fast cook-off	3.9	4.6.18	--

4.5.1.2.2 Type II. The panels of 4.5.1.2.1 requiring Type II topcoat for tests specified herein shall be coated by brushing to a dry film thickness of 20 +5 mils. The coating system shall be cured an additional 16 hours at 180 +2°F.

4.5.2 Application of the coating system to the LAU-10D/A rocket launcher specimens. Unless otherwise specified, preparation of the substrates

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shall be in accordance with 4.5.1 and 4.5.1.1. Equipment shall be as specified in 4.6.5 with the following dry film thickness criteria:

a. Epoxy polyamide primer (MIL-P-23377): 0.6 to 0.8 mils

b. Type I coating (air dry 4 hours prior to application of Type II coating): 160 ±14 mils

c. Type II coating (air dry a minimum of 24 hours prior to test): 10 mils minimum

d. Total dry film thickness of the coating system: 170 ±15 mils

4.6 Test methods.

4.6.1 Types I or II, Component I.

4.6.1.1 Viscosity. The viscosity shall be determined in accordance with Method 4287 of Fed. Test Method Std. No. 141. For Type I, Component I material, test at 120 ±2°F using a No. 7 spindle and a speed of 10 RPM. For Type II, Component I material, test at 73 ±2°F, using a No. 6 spindle and a speed of 10 RPM. Record the viscosity after 5 revolutions.

4.6.1.2 Solids content. Determine the solid content in accordance with Method 4041 of Fed. Test Method Std. No. 141 except for the following:

a. Form a slurry using a 1 gram sample of Type I, Component I and approximately 30 milliliters of chlorobenzene, or a 1 gram sample of Type II, Component I and approximately 30 milliliters of toluene.

b. Filter the solids from the slurry using glass fiber filters (Grade 934 AH, Reeve Angel or equivalent).

c. Oven dry filters at 55 ±2°C for one hour. Cool to room temperature and determine the solids content.

4.6.1.3 Weight per gallon. The weight per gallon of Type I or II, Component I material shall be determined in accordance with Method 4184 of Fed. Test Method Std. No. 141.

4.6.1.4 Epoxide equivalent. Separate the epoxy resin from Component I by centrifuging. Determine the epoxide equivalent in accordance with Method 7403 of Fed. Test Method Std. No. 141 except as follows:

a. Adjust the sample size to obtain a 10-15 ml. titration end point (Approximately 0.4 grams).

b. Use chlorobenzene in place of chloroform.

4.6.2 Type I or II, Component II.

4.6.2.1 Total amine value. The total amine value of Type I or II, Component II curing agents shall be determined in accordance with ASTM Method D 2073.

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4.6.2.2 Viscosity. The viscosity shall be determined in accordance with method 4287 of Fed. Test Method Std. 141. For Type I, Component II material, test at 150 ±2°F using a No. 3 spindle and a speed of 20 RPM. For Type II, Component II material, test at 77 ±2°F using a No. 4 spindle and a speed of 10 RPM.

4.6.2.3 Weight per gallon. The weight per gallon shall be determined as specified in 4.6.1.3.

4.6.3 Condition in container (Types I or II Components). Each component material shall be stirred and observed for smoothness, homogeneity, absence of pigment flotation, course particles or settling.

4.6.4 Storage stability (Types I & II Components). The unopened containers shall be placed in an area where they will not be agitated or disturbed for a period of 6 months at 65 ±25°F and then checked for conformance to Tables I, II, III, and IV as applicable.

4.6.4.1 Accelerated stability. A full unopened container each of Types I and II Component I material shall be stored in an oven at 120 ±4°F for a period of 6 hours and cooled to room temperature and checked for conformance to 3.4.6.1.

4.6.5 Application. Type I and Type II coatings shall be sprayed on panels prepared as specified in 4.6.6.1 with the following equipment: Binks Variable Ratio Formulator G, Model No. 101-64212 with Binks Model 18FA automated spray gun or equivalent.

4.6.6 Thermal efficiency.

4.6.6.1 Panel preparation. Prepare ten panels each 0.040 by 8 by 8 inches in size and six discs each 1-15/16 inch in diameter by 0.040 inch in thickness from aluminum (6061) sheets conforming to QQ-A-250/11, Condition T-6. Cut a 2-inch hole in the center of each panel and cement the discs in the holes with a metal-to-metal type epoxy bonding compound capable of withstanding high temperatures. Cementing the discs in the panels reduces the thermal conductance effects from the edges. The test panels shall be prepared as specified in 4.5.1 and 4.5.1.1. Type I material shall be spray-applied (see 4.6.5) and subsequently machined to a uniform film thickness of 175 ±5 mils. After machining, spray Type II topcoat to a dry film thickness of 20 ±5 mils. Save three panels for the humidity test of 4.6.14, and two panels for the weather resistance test, 4.6.16.

4.6.6.2 Procedure. This test exposes the coating to a fire environment where a total flux of 10 BTU/FT^2 - sec as measured by an asymptotic calorimeter is 90 percent radiative in a slightly fuel rich condition. The average emissivity of the source of the radiant heat flux shall be at least equal to 0.9. Aviation fuel (JP-4 or JP-5) conforming to MIL-T-5624 shall be used to fire the unit.

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The coated panel thickness shall be measured and recorded. When ready to test, the panel shall be inserted in a transite mask or any thermally insulated holder which allows the panel to rest in about 1/16 to 1/8 inch of the edges of the coated surface. The backside of the test panel shall be insulated with nominally 6 lbs/ft^3 refractory fiber insulation. At time, T = 0, the masked panel shall be placed over the fire box and a thermocouple in contact with the back of the disc records the temperature rise on the panel. The maximum initial panel temperature prior to fire testing shall be 90°F. The flux shall be uniform over the entire panel and the center disc shall be approximately 8-10 percent of the exposed area and shall be representative of a larger, semi-infinite panel. The time-temperature data and the film thickness in mils shall be recorded and a report made of the time, in seconds, to reach the temperature of 260°C (500°F) and 538°C (1000°F) for the recorded mil thickness of the coating. The test results shall conform to 3.6.1 and the data reported shall be an average of three determinations.

4.6.7 Tensile properties. The tensile properties shall be determined in accordance with Method 1011, Fed. Test Method Std. No. 406. The test specimen shall be Type I, 1/4-inch maximum sheet, Figure 1011A, 0.125 inch thick. Test specimens shall be conditioned for 16 hours at 180 ±2°F and cooled to room temperature prior to testing.

4.6.8 Color, Type II. Determine the color of a coated panel in accordance with the photoelectric tristimulus data Method 4252 of Fed. Test Method Std. No. 141. The gloss shall be compared with the specified color of Fed. Std. 595.

4.6.9 Hardness (Type I). Mix Components I and II at the ratio specified in 3.2.1 and prepare a sample casting 4 inches in diameter by 1/2 inch thick. Cure for a minimum of 2 hours at 150°F. Cool sample to room temperature and test for hardness in accordance with Method 1082 of Fed. Test Method Std. No. 406.

4.6.10 Adhesion. Panels shall be aluminum alloy conforming to QQ-A-250/5, T-0 Condition, and anodized in accordance with MIL-A-8625, Type II. They shall be 0.020 by 5 by 12 inches in size and finished with epoxy primer, and Type I and II coating compounds specified in 4.5.1.1, 4.5.1.2.1, and 4.5.1.2.2. The panels shall be tested in accordance with Method 6221 of Fed. Test Method Std. No. 141 and checked for conformance to 3.6.5.

4.6.11 Lubricating oil resistance. Panels shall be prepared as specified in 4.5.1 through 4.5.1.2.2. The panels shall then be immersed in lubricating oil composed of 95 percent of di-2 ethyl-hexyl sebacate, 1/ and 5 percent tricresyl phosphate, by weight, at a temperature of 250 ±4°F for a period of 4 hours. Four hours after removal, the panels shall be examined for conformance to 3.7.1.

1/ di-2 ethyl hexyl sebacate may be obtained commercially as "Plexol 201" from Rohm and Haas.

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4.6.12 Fungus resistance. Panels shall be prepared as specified in 4.5.1 through 4.5.1.2.2 and tested for resistance to fungus in accordance with Method 508, Procedure I of MIL-STD-810. Upon completion of the fungus test check for conformance to 3.7.2

4.6.13 Impact resistance.

4.6.13.1 Panel preparation. Panels shall be aluminum alloy conforming to QQ-A-250/5, T-0 Condition and anodized in accordance with MIL-A-8625, Type II. They shall be 0.020 by 5 by 12 inches in size, and finished as follows: Apply epoxy primer and Type I and II coatings as specified in 4.5.1.1 through 4.5.1.2.2, except that the Type I coating thickness shall be 0.160 ±0.015 inch and the Type II coating shall be 0.015 ±0.005 inches.

4.6.13.2 Impact test. Place one prepared panel from 4.6.13.1, film side up over the hole of the cylindrical base of a Gardner Heavy Duty Impact Tester 1G-1120. Determine the maximum impact resistance in inch-pounds at which no failure occurs. Repeat the test on the reverse side of another panel from 4.6.13.1 with the film side down. Check for conformance to 3.7.3.

4.6.14 Humidity resistance. Three panels prepared as specified in 4.6.6.1 shall be subjected to 336 hours of exposure at 100 percent relative humidity at 120°F in accordance with the method specified in ASTM D 1748.

4.6.15 Resistance to temperature shock. Three panels prepared as specified in 4.5.1 through 4.5.1.2.2 shall be tested in accordance with Method 503 of MIL-STD-810B except as follows:

a. The internal hot chamber shall be maintained at 140°F and 95 percent RH.

b. The internal cold chamber shall be -40°F.

c. Panels shall be cycled every 24 hours in lieu of 4.

d. The complete test period shall be 28 days.

4.6.16 Weather resistance. Four panels shall be prepared as specified in 4.6.6.1 and exposed outdoors for one year in the vicinity of Miami, Florida in accordance with ASTM Test Method D 1014. The panels shall be mounted at 45 degrees facing south. After exposure, the panels shall be checked for conformance to 3.7.6 and subjected to the thermal efficiency test.

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4.6.17 Environmental performance tests. The coated LAU-10D/A launchers of 3.8 shall be subjected to the tests of 4.6.17.1 through 4.6.17.3 and checked for conformance to 3.8.1.

4.6.17.1 Temperature and humidity cycling. Launchers shall be in the up shipping configuration when tested. Testing shall be in accordance with WR-50 except that the temperatures shall be modified to +140°F (95% RH) and -40°F.

4.6.17.2 Transportation vibration and abrasion test. Launchers shall be in the up shipping configuration when tested. Testing shall be in accordance with Method 514.1, Figures 514.1-7 and 514.1-9, Curve (AV:AW), and time schedules specified for Procedure X as shown in Table 514.1-VII of MIL-STD-810. The launchers shall be tested at standard ambient conditions (MIL-STD-810), +140°F and -40°F, in that order.

4.6.17.3 Aircraft vibration test. The launchers shall be in the flight configuration and tested at standard ambient conditions (MIL-STD-810), and at -40°F. The test procedure and time schedule shall be in accordance with Method 514.1, Table 514.1-II and Curve H of Figure 514.1-2 of MIL-STD-810.

4.6.18 Fast cook-off. The fast cook-off test shall be conducted in accordance with Naval Weapons Requirement WR-50 under the supervision of the test facility. In addition, the following shall apply:

a. The quantity of fuel shall be sufficient to cause reaction of the ordnance or to insure a 15-minute fire. Test fuel shall be JP-4 or JP-5 conforming to MIL-T-5624.

b. The test pit shall be a minimum of 20 feet by 20 feet.

c. Flame temperature to the launcher shall reach 1000°F within 30 seconds after fire ignition.

d. The launcher shall be suspended from its suspension lugs with its center 30 to 40 inches above the fuel.

e. The test shall not be performed when the wind velocity exceeds 5 knots.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Type I and Type II, Components I and II unit packages. Components I and II shall be packaged in pails conforming to PPP-P-704, Type II, Class 3; or Type II, Class 3 or 4; in drums conforming to PPP-D-729, Type III or IV, as specified by the procuring activity (see 6.2). The

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inside surface of the pail or drum shall be treated as required to preclude content contamination.

5.1.2 Level C. Coating compound components shall be packaged in unit capacity containers in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be Level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Type I and Type II, Components I and II. Components I and II packaged in pails as specified in 5.1.1.1 shall be palletized in accordance with MIL-STD-147.

5.2.2 Level C. Material packaged as specified in 5.1.2 shall be packed in such a manner that will insure acceptance by common carrier at lowest rates and safe delivery at destination. Shipping containers shall comply to the Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. Marking for shipment shall be in accordance with MIL-STD-129 and shall include the following nomenclature and information:

Coating Identification: Type I Base Coat or II Topcoat, as applicable
Component Identification: Component I or II, as applicable
Manufacturer's Name or Trademark
Date Manufactured, Month and Year
Lot Number and Batch Number
Net Contents
Specification MIL-C-81946(AS)
Flash Point (In accordance with ICC Regulations)

5.3.1 Additional marking. Individual containers shall be marked with the following caution label:

"Caution:
Avoid contact with skin and eyes. Keep containers closed. Avoid prolonged or repeated breathing of vapor. Use with adequate ventilation. Wear fresh clothing daily when contacting Component II Curing Agent.
Storage temperature: 40 to 90°F."

5.3.1.1 Application instructions. In addition to the labels specified in 5.3.1, each component unit shall contain a printed label with the following instructions:

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"Application instructions: This is a two coating system consisting of Type I base coat and Type II topcoat. Type I coating consists of two components and Type II coating consists of two components. To apply Type I coating, mix 1 part by weight of Component II to 4 parts by weight of Component I. For Type II coating, mix 1 part by weight of Component II to 9 parts by weight of Component I."

6. NOTES

6.1 Intended use. The thermal insulating coating system is intended for use as an exterior protective coating on aircraft ordnance items. This coating is formulated to meet applicable environmental conditions and also to delay an explosive reaction when the coated ordnance item is exposed to high flux thermal environments.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Type I or II (see 1.2).

c. Components (see 3.4.1 through 3.4.4).

d. Level of packaging and packing (see Section 5).

e. Quantity in pounds (avoirdupois)

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Requests for information pertaining to qualification of products covered by this specification may be obtained from the Commanding Officer, U.S. Naval Air Development Center, Johnsville, Warminster, Pennsylvania 18974, Attention: Code 30223.

6.4 Supersession data. This specification supersedes Naval Air Systems Command Purchase Description AS-3477 dated 2 October 1972.

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6.5 Batch. A batch is defined as the quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-C-81946 (AS) COATING SYSTEM, THERMAL INSULATION, INTUMESCENT

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO (If "Yes", in what way?)

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